

5G APPLICATIONS

INDUSTRIES 4.0

Predictive Maintenance:

System engineers can predict and anticipate problems and take care of them before they happen, through the use of smart sensors.

Hazardous Process:

From assembly lines to manipulating cranes, workers will be able to remotely operate equipment, reducing the risk to human life.

Connected Goods:

With low cost chips enabling smart connected things (SCoT), the capabilities of a product can extend beyond its physical presence.

Inventory Management:

Sensors can manage the inventory better, and respond to demand to increase production or reduce it as per requirements.

Cyber-Physical systems:

The high quality connectivity will allow workers in remote locations to manipulate objects in real time through robotics and VR



TRANSPORT

Smart Parking:

Using cameras and vehicle counters, smart parking solutions can save fuel, time and space, by parking vehicles autonomously.

Traffic Alleviation:

Citywide response from locations such as the Fire Brigade or hospitals can clear routes when necessary, for the use of emergency vehicles.

Truck Platooning:

A convoy of autonomous trucks driving at near constant speeds constantly coordinating with each other can improve fuel efficiency.

Toll Auto Payments:

Although available in some places, the low cost 5G chips will allow automatic payment of tolls throughout highways.

Self Driving Cars:

Self driving cars can coordinate with each other, with the network supporting the real time reactions needed for safety.



MEDIA

Cloud Based Gaming:

The reduced latency will finally support multiplayer gaming based out of cloud platforms, freeing users from the hardware.

UHD Video:

The most basic of the multimedia options in the future, high resolution videos upwards of 8k can be streamed without buffering.

3D Video:

You can experience music concerts and sporting events in realtime, from the best seats in the stadium, no matter where in the world you are.

Virtual Reality:

Long VR sessions will be possible because the low latency means that the virtual environment reacts in real time to user movement.

Augmented Reality:

Head mounted displays can be used without cables, as most of the heavy processing can take place in the cloud.



SECURITY

End to End secured slices:

The network can be sliced up to provide different levels of reliability and security according to the requirements of the application.

Identity Management:

Identity and access management is built into the network itself, supporting all the devices for all the users.

IoT Security:

Security for IoT devices can be supported by the network itself, instead of being relegated to each manufacturer and platform.

Personal body cameras:

Used by some law enforcement departments around the world, personal body cameras will become cheaper for everyone.

360 degree Video Surveillance:

There are already 5G surveillance cameras on the market, which allow real time 360 degree monitoring in high resolution.